

JavaScript Revision Assignment

Topics Covered: Variables, Data Types, Mathematical Operators, Comparison Operators, and Conditional Statements.

Solve the following problems using JavaScript.

Problem 1: Student Result Checker

Create variables for `studentName`, `marksObtained`, and `totalMarks`. Calculate percentage. If percentage is 90 or above, print 'Excellent'; 80-89 'Very Good'; 70-79 'Good'; 60-69 'Average'; below 60 'Needs Improvement'.

Problem 2: Age Eligibility System

Create variables `age` and `hasValidID`. Using comparison operators and conditional statements, check whether a person is eligible to enter an event. A person can enter only if `age` is 18 or above and `hasValidID` is true.

Problem 3: Electricity Bill Calculator

Create a variable `unitsConsumed`. Calculate the bill as follows: If units are less than or equal to 100, rate = ₹5/unit; 101-300, rate = ₹7/unit; above 300, rate = ₹10/unit. Print total bill amount.

Problem 4: Largest of Three Numbers

Create three variables `num1`, `num2`, and `num3`. Using comparison operators and conditional statements, find and print the largest number. Also handle the case when all numbers are equal.

Problem 5: Online Shopping Discount Challenge

Create variables `purchaseAmount` and `isPremiumMember`. Apply discounts: Premium members get 20% discount if purchase is above ₹5000, otherwise 10%. Non-premium members get 15% discount if purchase is above ₹7000, otherwise 5%. Print original amount, discount amount, and final payable amount.